MediXAI

Multimodal AI for Prescription Analysis and Personalized Health Monitoring

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Presentation Outline

Introduction

MediXAI Overview

MediXAI is a multimodal healthcare framework for:

- Automated prescription image analysis (handwritten and image)
- Personalized health monitoring using patient signals and records
- Supporting clinical decision-making systems

Why it matters?

- Prescription misinterpretation leads to medical risks
- Existing systems lack multimodal integration
- Demand for AI-based personalized healthcare is increasing

Key Idea:

- Fusion of image, text, and patient data for intelligent healthcare prediction

Problem Statement

Major Problem

- Existing systems are single-modal (OCR / text / image only)
- No unified multimodal healthcare system exists

Key Limitations

- High error in real-world prescriptions
- Poor performance on noisy / handwritten data
- No real-time clinical deployment system

Motivation

- Prescription misreading causes medication errors
- Patients & pharmacists often misunderstand handwritten notes
- Need AI-based safe prescription interpretation system

Objectives of the Review

- **Model Comparison:** ML/DL, CNN, XAI, OCR, NLP Vision-Language models
- **Metrics:** Accuracy, Precision, Recall, F1-score, latency
- **Robustness:** Blur, noise, low-quality scans, handwriting variation, multilingual data
- **Limitations:** Data scarcity, weak validation, poor generalization, clean-data dependency
- **Explainability:** LIME, SHAP, Grad-CAM

Research Gap: Separate OCR-based prescription analysis and health monitoring systems; no unified end-to-end clinical AI framework.

Paper Selection Methodology

- **Total Papers Reviewed:** 21 research papers
- **Sources:**
  - IEEE Xplore
  - Google Scholar
  - Kaggle
  - ScienceDirect
  - ResearchGate
- **Publication Years:** 2020 – 2026
- **Inclusion Criteria:**
  - Medical OCR, prescription analysis, or healthcare AI related studies
  - ML, DL, multimodal AI, or XAI-based approaches
- **Exclusion Criteria:**
  - Non-medical domain papers
  - Non-AI based traditional rule-based systems

Literature Review

Category	Methods / Papers	Key Findings
OCR + NLP Pipeline	OCR + NLP (ViT, BERT), error propagation analysis	Good extraction (71–84% accuracy)
CNN + CRNN OCR Systems	CNN + BiLSTM + CTC loss	70–82% accuracy, affected by imbalance and handwriting variation
Hybrid OCR + NLP	CNN-LSTM + Tesseract OCR + NLP	Around 92% accuracy, but fails on low-quality images
Hybrid OCR Pipelines	ANN-based OCR, fax prescription dataset	CRR 94.5%, but limited generalization
RAG + OCR Systems	OCR + NLP + RAG models	92% accuracy, but needs human verification
YOLO + OCR	YOLO + OCR + Spell Correction	High ROI accuracy (99%)

Key Insight: Most systems show good accuracy, but real-world clinical deployment is still limited due to data and validation issues.

Comparative Analysis

P	Method	Accuracy	Advantages	Limitations
P1	OCR + NLP Pipeline	71–84%	Good system-level analysis of OCR errors	No real clinical dataset
P2	CNN + CRNN OCR	70–82%	Works for basic handwritten recognition	Sensitive to imbalance and handwriting style
P3	CNN-LSTM + Tesseract OCR	92%	Good medical entity extraction	Weak on poor-quality images
P4	MIRAGE (LLaVA + Qwen-VL + CLIP)	82% extraction	Strong multimodal approach	Vision encoder bottleneck
P5	OCR + NLP + RAG System	92%	Structured medical outputs	LLM hallucination issues
P6	YOLO + OCR + Spell Correction	96–99% ROI	Very high detection accuracy	No full end-to-end evaluation
P7	Donut (OCR-free Transformer)	84–88%	End-to-end document understanding	Limited dataset and validation

Key Insight: Most methods perform well on specific tasks, but real-world medical prescription understanding is still unreliable.

Research Trends and Observations

Popular Methods

- CNN, CNN-LSTM, DONUT, YOLO
- OCR + NLP, RAG, Multimodal AI

Performance Trend

- CNN/RNN: 80–82%
- CNN-LSTM: $\sim$92%
- YOLO: 99.6% (ROI)

Key Observations

- Hybrid systems give better real-world performance
- Multimodal models are improving understanding
- XAI is used for trust and transparency

Final Insight: YOLO performs best for detection, while CNN-LSTM and hybrid OCR+NLP models are most practical for end-to-end medical document analysis.

Research Gaps

- **Data Limitation** Small and synthetic datasets reduce generalization.
- **Deployment Gap** Models are not tested in real clinical environments.
- **Noise Issue** Performance drops on blurred and handwritten prescriptions.
- **Integration Gap** No unified OCR, NLP, AI, validation system.
- **Clinical Validation** Limited real hospital or pharmacy testing.
- **Interpretability** XAI is still not fully reliable for medical decisions.

Final Insight: Need for robust, real-time, clinically validated healthcare AI systems.

Future Research Directions

- **Edge AI & Real-Time Systems** Lightweight models for fast on-device prescription processing.
- **Federated Learning** Privacy-preserving training without central medical data storage.
- **Improved Vision Encoders** Better handwriting understanding using SigLIP / InternVL.
- **Advanced OCR Models** Adoption of TrOCR, Donut for robust document extraction.
- **Multimodal Healthcare AI** Integration of OCR, NLP, and clinical decision support systems.
- **Explainable AI Enhancement** More reliable XAI using hybrid approaches (LIM, SHAP, CAM).

Conclusion

- **Overall Observation** Multimodal OCR systems are effective for medical document understanding but still face real-world challenges.
- **Best Approaches** Vision-language models with improved encoders (SigLIP, InternVL) and OCR models like DONUT and TrOCR perform better than traditional methods.
- **Strengths** Good performance on structured data and strong automation capability.
- **Limitations** Weak performance on noisy handwriting, limited datasets, and lack of clinical validation.
- **Final Remark** Further improvements are required for reliable real-world healthcare deployment.

Team Contribution

Team Member ID	Papers Reviewed	Report Contribution	Presentation Contribution
23201009	7 Papers	Introduction, Problem Statement, Objectives, Methodology Literature Review	Prepared Intro, Methodology and Literature Review Slides
23201023	7 Papers	Comparison, Research Gaps, Trends	Prepared Analysis and Comparison Slides
23201050	7 Papers	Future Work, Conclusion, Discussion, References	Prepared Final Slides and Editing

References

- Mankash et al., "MIRAGE: Multimodal Identification and Recognition of Annotations in Indian General Prescriptions," arXiv, 2024.
- Daniela Gifua, “AI-backed OCR in Healthcare,” None, 2023.
- Shaik Sharjeel and Mohammad Ubaidulla Arif, “AI-Based OCR System for Handwritten Medical Prescription Recognition and Interpretation,” None, 2025.
- Hung Truong Thanh Nguyen et al., “ Evaluation of XAI: SHAP, LIME, and CAM ” Conference Paper, 2021.

Thank You

Questions?